

Does a feudal reinforcement learning improve recovery time following abrupt opponent strategy shifts, compared to flat RL approach (PPO)?

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Abstract

Reinforcement learning is increasingly applied to competitive settings in which an opponent’s behaviour is not fixed. When that behaviour changes abruptly, the resulting non-stationarity can invalidate a previously effective policy. Hierarchical architectures are widely expected to re-adapt faster than flat ones in such situations, yet this expectation has rarely been tested directly against a shifting opponent. This work investigates whether a hierarchical Feudal Network (FuN) recovers more quickly than a flat Proximal Policy Optimization (PPO) agent following abrupt opponent strategy shifts in the game of Go. Both agents are trained against fixed heuristic opponents in PettingZoo’s go_v5 on 9×9 and 13×13 boards, after which mid-training opponent shifts of varying magnitude and frequency are introduced. Their effect is measured by recovery time, together with the depth and overall cost of the resulting performance drop, over multiple random seeds. Across the conditions examined, the hierarchical agent does not recover faster than the flat baseline. These findings suggest that, the FuN hierarchy does not provide the expected advantage under opponent-induced non-stationarity.

1 Introduction

- Motivate: real opponents change strategy mid-interaction - the agent’s policy suddenly underperforms and must re-adapt (non-stationarity).
- The idea under test: a manager/worker hierarchy should recover faster.

1.1 Research Question

- State RQ3 exactly. Describe the setting: the game of Go, against fixed rule-based heuristic opponents on two board sizes (9×9 and 13×13).
- What this work adds: (i) a controlled opponent-shift recovery benchmark in Go; (ii) a tested FuN vs PPO comparison under identical conditions, across shift magnitude, frequency and board size, (iii) three recovery measures and what each can and cannot detect.
- Main finding: the hierarchy did not improve recovery.
- Roadmap: preview the remaining sections - Related Works, Design and Implementation, Results, Discussion, and Conclusions.

2 Related Works

- Open with: flat versus hierarchical RL, both of which must handle a shifting opponent - followed by a preview of sub-sections.
- Well-known RL work on Go, such as AlphaGo, as background context.

2.1 Non-stationarity in Multi-Agent RL

- Non-stationarity / opponent-induced dynamics in multi-agent RL; why a shift breaks a learned policy.

2.2 Flat and Hierarchical RL

- Flat on-policy RL: PPO - the baseline under test.
- Hierarchical RL and Feudal Networks: Manager-Worker, temporal abstraction, intrinsic sub-goal reward; note the claims that it adapts to a different opponent.

2.3 Measuring Adaptation and Recovery

- Recovery-time / adaptation-speed metrics; how the recovery measure relates and contrasts with catastrophic-forgetting work.
- State the gap this paper fills: HRL vs flat recovery under an opponent shift, tested directly.

3 Design and Implementation

- Open with: brief outline of the section - the environment and opponents, the agents, the experimental design (Phase 1 and Phase 2), the metrics, and the tools used.

3.1 Environment and Opponents

- **Environment:** PettingZoo go_v5; 9×9 (komi 5.5) and 13×13 (komi 7.5); area scoring. Note the move-cap / forced-termination handling.
- **Fixed opponents:** five heuristic bots (greedy, defensive, corner, edge, random), description of each style.

3.2 Agents

- **Agents:** PPO (action-masked categorical policy, shared CNN backbone); FuN (matched CNN perception, Manager dilated-LSTM trained on extrinsic reward, Worker trained on intrinsic cosine reward). State that both share observation preprocessing and backbone so the comparison is

architectural; note key implementation corrections made to FuN.

3.3 Experimental Design

- **External reference:** GNU Go, to reference agent strength before the comparison.
- **Phase 1:** train each agent against each opponent to a stable win-rate baseline; this baseline defines the recovery target.
- **Phase 2:** resume a Phase-1 checkpoint, then apply abrupt opponent shifts.
 - Magnitudes, calibrated by observed disruption: LOW corner→edge, MED corner→defensive, HIGH greedy→defensive
 - Frequencies: f1 single shift, f2 periodic, f3 frequent.
 - Design table: 3 magnitudes $\times$ 3 frequencies $\times$ 2 agents $\times$ 2 boards $\times$ 3 seeds.
- **Table of experiments:** summarise the runs in a table - columns for condition, what is measured, and purpose.
- **Parameters:** exact shift schedules and step budgets.

3.4 Metrics

- **Metrics used:** Rolling win rate (window 100 games); recovery time; dip depth; adaptation cost. Description and purpose of each metric.

3.5 Tools

- **Setup:** GPU and CPU training on the UCT cluster and CPU training on the CHPC cluster; metrics logged to Weights & Biases and to per-game CSV files; per-seed runs kept separate then merged; state the number of seeds.

4 Results

- Open with: brief outline of the results - Phase-1 capability baselines, then Phase-2 recovery, presented as tables and figures.

4.1 Phase-1 Capability

- **Phase-1 capability** - Table: win rate per agent $\times$ opponent $\times$ board, with the GNU Go reference [values TBD].

4.2 Phase-2 Recovery

- **Phase-2 recovery** - Table: recovery time, dip depth and adaptation cost per agent $\times$ magnitude $\times$ board, as mean over seeds [values TBD].
- **Figures:**
 - rolling win-rate curves with shift points marked, PPO and FuN overlaid per condition;
 - grouped bar chart of recovery time by magnitude, per board;
 - dip-depth / adaptation-cost comparison across conditions.

5 Discussion

- Open with: recap what was tested, then interpret the Results.

- Does the result answer RQ3 / reject the hypothesis that hierarchy speeds recovery?
- Analyse *why* the result is negative - candidate mechanisms.
- How recovery depends on magnitude, frequency and board size, and what the two boards reveal about the metrics themselves (recovery time is informative on 9×9 ; dip depth / adaptation cost on 13×13).
- How the finding aligns with or contrasts the related literature (HRL's claimed recovery advantage vs an opponent-shift setting).
- **Implications:** what the negative result means for using HRL under opponent-induced non-stationarity, and when a hierarchy might still help.
- **Limitations:** all seeds start from the same Phase-1 model, so they capture only Phase-2 randomness; win rate used rather than a territory/score margin; boards limited to 9×9 and 13×13 , not full-size 19×19 Go.

6 Conclusions

- Restate the answer to RQ3 in one line [direction known, final numbers TBD]: the FuN hierarchy did not recover faster than flat PPO.
- Implication for hierarchical RL under opponent-induced non-stationarity.
- Future work: more seeds; 19×19 board; a territory-margin metric.